Use of the material, and why they are suitable for the following product **(6 marks)**

1. MDF – Flat Pack Furniture
2. Concrete – Paving Slabs
3. Carbon Fibre Reinforced Plastic – Fishing Rods

# MDF – Flat Pack Furniture

MDF is suitable for use in flat pack furniture as it is available in large sheets/sizes; this means that the product can be assembled easier in fewer pieces and is simpler to join compared to other woods limited in size sourced from trees e.g. Oak and Mahogany. MDF can also be recycled and as flat pack furniture typically tends to have a lower product use life, this is beneficial for the environment as it conserves trees and other material making it sustainable source. MDF tends to be cheaper than tree-sourced woods such as Ash and as a result, this is cost effective to manufacture and the savings may then be passed onto the consumer as it is more affordable and cheap. MDF can be finished in a range of coats e.g. paint, gloss or veneers. This gives the impression of a more premium feeling and expensive product – this will prove to be aesthetically pleasing to the consumer making it a more appealing product to purchase. In terms of load bearing capabilities and compressive strength, MDF is very good material for us in the majority of flat pack furniture such as tables or cupboards which tend to bear weight e.g. items placed on a shelf or a work surface. Using MDF in these products will therefore mean that they will hold structural rigidity and form a lot better than alternative woods as it won't bend or bow under strenuous loads.

## Concrete – Paving Slabs

Concrete is a useful material in the production of paving slabs because it can resist compression and a lot of wear on the slab itself; people walk over the surface the majority of time and the good compression load capabilities mean that it won't deform and lose it's shape. Concrete paving slabs are also significantly cheaper than other materials like tiles or slate which means that it is cost effective and can be used more widely in a range of different environments e.g. public walkways or construction sites. The paving slabs can be made in a variety of sizes (almost any economically viable size) if the concrete is mixed and poured there and then on the construction site meaning that it is quite a versatile material to work with not restricted by size or shape constraints so the slabs can be used to fit almost any area. When orders are placed, delivery of concrete paving slabs can be done very rapidly and as a result, building work can be completed quickly and efficiently as well as this, colours can then be added and mixed to during production which will offer a better aesthetic appeal to the customer using the paving slabs

**Carbon Fibre Reinforced Plastic – Fishing Rods**

Using Carbon Fibre reinforced plastic in fishing rods has many advantages. Firstly, it is very lightweight when compared to steel or aluminium meaning that it will be easier for the customer to hold and use for extended periods of time. It is an extremely strong material and can resist high loads which, when fishing, the rod will be subjected to when catching fish. This strain will be repeated multiple times over the course of the lifetime of the fishing rod and it is vital that it can cope with strenuous loads. Because of it's hard wearing capability, it can last for a very long time and the fishing rod will be long lasting and durable. This will benefit the consumer and also the environment as it reduces waste in landfills and the product will last longer without warping or deforming. Carbon Fibre reinforced plastic is also resistant to chemical/biological attack which in fishing is important as it is exposed to seawater (which can corrode/rust steel) or bait which may leave residue on the fishing rod but as it is resistant to this, it won't wear down and lose structural integrity. It is a versatile material and a lot of shapes can be produced meaning that new designs of fishing rods can be easily manufactured and produced which will lead to stronger designs and revolutionary new mechanisms leading to more satisfied customers.